

One Layer at once

Below is a **deep-dive “question catalogue”**—the *user-facing* queries that Arth should be able to answer **inside each layer, without cross-referencing any other layer**.

Use this to reverse-engineer data models, API payloads, calculation engines, alert rules, and UX surfaces.

Layer 0 — Financial-OS Governance & Improvement

1. Data-pipes & refresh integrity

- Which institution–API connections failed or returned partial data in the last 24 h?
- How many transactions were auto-categorised with <80 % confidence this week?
- Which dashboards are > 12 h stale relative to their SLA refresh cadence?

2. Automation rules & playbooks

- Which payday sweep, bill alert, or portfolio-rebalance rule fired since the last daily run?
- Were any automations skipped because of missing data or breached guard-rails?
- Which rules are queued for the next cron cycle but lack the required approvals/limits?

3. DevSecOps hygiene

- When was the last 2-FA audit for every connected account?
- Which API keys, passwords, or OAuth tokens expire within 30 days?
- Have any new budgeting or brokerage apps been added that still need security vetting?

4. Metric library upkeep

- Which KPIs were modified (formula, threshold, owner) this quarter?

- Are any definitions duplicated or unused for > 90 days?
- Do any metrics lack unit tests / back-tests across historical data?

5. Continuous-improvement backlog

- What open “tech-debt” tasks are overdue (e.g., migrate to v2 of a vendor API)?
 - Which suggested new metrics/automations are awaiting prioritisation?
 - Did the last monthly retro create action-items still unassigned?
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Layer 1 — Net-Worth Snapshot (Balance-Sheet View)

1. Point-in-time inventory

- What is our total net worth as of *right now*?
- Break down assets and liabilities by category, sub-category, and individual holding.
- Which five positions contribute the most to net-worth volatility?

2. Valuation & data lineage

- What pricing sources (NSE, BSE, Mutual-Fund NAV, Zillow, manual) fed each value?
- When was each valuation last refreshed, and what is the data lag (minutes/hours/days)?
- For private assets (ESOPs, collectibles), which valuation method and assumptions were used?

3. Concentration & exposure diagnostics

- What % of net worth sits in employer equity (vested + unvested)?
- How much real-estate exposure do we have relative to liquid assets?

- What is the INR vs. foreign-currency split across the portfolio?

4. Movement analytics

- How has net worth changed Day-over-Day, MoM, and YoY?
- Which 3 assets created ≥ 80 % of the net-worth change in the last quarter?
- Which liabilities amortised faster/slower than schedule?

5. Stress-test queries

- If equities drop 25 % tomorrow, what is the simulated new net worth?
- What happens to LTV ratios on secured loans if property valuations fall 10 %?

6. Prescriptive prompts

- Which single liability pre-payment gives the biggest increase in net worth per ₹1 lakh?
- Which low-yield asset should be reallocated first to meet target asset allocation?

Layer 2 — Earnings & Spending Profile (Income / P&L)

1. Income diagnostics

- List all active income streams with gross, net, and variance vs. last period.
- What % of total comp is fixed salary vs. variable vs. equity?
- Which side-hustle or dividend source has the highest YoY growth?

2. Expense forensics

- Rank expense categories by share of total spend and YoY growth.

- Identify subscriptions with cost $\uparrow > 15\%$ since last renewal.
- What was the single biggest discretionary purchase this quarter?

3. Surplus & efficiency metrics

- Rolling 3-, 6-, 12-month savings-rate trends and confidence intervals.
- What is the break-even salary figure if current discretionary spend $\uparrow 10\%$?
- How many months in the past year showed a negative operating surplus?

4. Behavioural insights

- Which “lifestyle creep” flags fired (e.g., dining-out $\uparrow >$ twice inflation)?
- Did any irregular bonuses get entirely spent within 30 days of receipt?
- Are there spend spikes around specific dates (e.g., travel, festivals)?

5. Scenario & planning

- How would taking a 20 % pay-cut impact annual surplus?
- What is the income required to sustain ₹X/month child-care costs from 2027 onward?

Layer 3 — Cash-Flow & Liquidity Dynamics

1. Day-to-day liquidity

- Current “cash runway”: how many months of core expenses can existing cash cover?
- What is the lowest projected bank balance in the next 30 days? On which date?
- How much can be liquidated within 1 bus-day, 3 days, 7 days?

2. Cash-flow calendar

- Visualise all expected inflows/outflows for the next 12 months (salary, EMIs, insurance, tuition, vacations, tax).
- Which month has the highest net outflow and by how much?
- Do any planned investments (SIPs, lump-sums) collide with large expense months?

3. Liquidity stress-tests

- If bonus pays *one month late*, does liquidity fall below 3-month threshold?
- In a double-EMI shock (interest-rate reset), how many days until the first cash-shortfall?
- How does selling RSUs vs. borrowing on a credit-line affect liquidity buffer?

4. Optimisation levers

- Which idle cash pockets exceed target minimums and can be swept into MMFs/T-bills?
- Is there excess credit-card headroom that could be converted to cheaper LOC?
- What staggered fixed-deposit ladder minimises reinvestment risk yet keeps 3-month access?

Layer 4 — Risk Protection & Contingent Exposures

1. Insurance adequacy

- Required vs. actual cover for life, term, health, disability, critical-illness. Gap (₹).
- How does sum-assured compare to 10× annual expenses or 15× annual income rule-of-thumb?
- Are any policies nearing renewal without auto-debit set up?

2. Emergency-fund health

- Fund size vs. target (6-, 9-, 12-month variants) and instrument mix (FD, liquid MF).
- Average time-to-cash for each emergency-fund component.
- How many times has the fund been tapped in the past three years?

3. Debt & covenant monitoring

- Debt-service-coverage ratio trend vs. lender covenants.
- Which loans have floating rates with reset dates in the next 6 months?
- What is the incremental EMI if rates rise 1 %?

4. Off-balance-sheet / tail risks

- Quantify parental care obligations (₹/year over next 5 years).
- RSU exercise obligations if employment ends today.
- Currency exposure from USD-denominated assets vs. INR liabilities.

5. Scenario analytics

- What happens to coverage adequacy if Koop-zee takes a 1-year unpaid sabbatical?
- How would a ₹10 lakh medical claim be funded (policy, emergency fund, asset sale)?

6. Mitigation roadmap

- Which single insurance top-up gives the greatest risk-coverage per premium rupee?
- What reserve should be earmarked for potential IVF costs in 2026?

Layer 5 — Tax & Legal Efficiency

1. Current-year tax posture

- Projected taxable income, deductions, and average vs. marginal tax rates.
- Section-wise utilisation: 80C/CCD, 80D, HRA, capital-gain exemptions.
- Outstanding advance-tax/withholding requirements and due dates.

2. Capital-gain strategy

- Unrealised gains/losses by holding period (short vs. long term).
- How much loss harvesting headroom remains this FY?
- Optimal sell-amount for RSUs to stay within the next lower surcharge bracket.

3. Entity & ownership optimisation

- Would using an HUF or LLP reduce tax drag on rental or consultancy income?
- Are property/co-ownership splits aligned with intended estate distribution?
- Do any foreign income streams trigger DTAA considerations?

4. Estate-plan completeness

- Status of wills, nominations, PoA for every major account.
- Are there any beneficiaries under 18 requiring a trust structure?
- When was each document last reviewed against changing assets/family situation?

5. Compliance & risk flags

- Have any crypto transactions exceeded the annual tax-reporting threshold?
- Do PF / NPS contributions breach upper limits that trigger excess-tax problems?
- Are any notarised documents near expiry (e.g., PoA for property sale abroad)?

6. Forward-looking optimisation

- How much additional voluntary EPF/NPS contribution maximises deduction without starving liquidity?
 - Should we defer RSU sale to the next FY to avoid surcharge creep?
 - What is the tax-efficient order of asset liquidation to fund a 2027 home down-payment?
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How to bring this to life

1. **Schema design** – Each bullet maps to *data attributes* (tables: transactions, holdings, policies, workflows).
2. **API connectors** – Identify required endpoints (banking AA, insurer APIs, CBDT tax portal, valuation feeds).
3. **Agent skill-set** – Descriptive reports, anomaly detection, scenario simulation, recommendation engine.
4. **UX surfacing** – Conversational intents (“show me...”, “simulate...”), dashboards, scheduled alerts.
5. **Layer-0 automations** – Guard-rail thresholds (e.g., cash runway < 3 months) that auto-trigger advice or sweeps.

Use this catalogue as the **single-source-of-truth for user-intent coverage** while you prototype databases, write OpenAPI specs, or craft LangChain toolkits. Once you start wiring data in, we can iterate on the answers, but for now this list ensures no critical question gets left unsupported.

Two Layers at once

Below is a **pair-by-pair question catalogue**—the *core user questions* that arise **only when two layers are viewed together** (Layers 1-5).

Treat each bullet as a *functional requirement* for Arth's query engine, calculators, alerts, and dashboards.

(Level 0—governance—is excluded as requested.)

Legend

- **L1** Net-Worth Snapshot
 - **L2** Earnings & Spending Profile
 - **L3** Cash-Flow & Liquidity Dynamics
 - **L4** Risk Protection & Contingent Exposures
 - **L5** Tax & Legal Efficiency
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1. L1 ↔ L2 (Net-Worth × Income/Expense)

Theme	Key questions
Conversion efficiency	<i>How much of last year's ₹-surplus actually shows up as net-worth growth? Which expense cuts/free-ups would raise net worth fastest?</i>
Source–use mapping	<i>Which income streams are earmarked for which asset classes (salary → SIPs, RSU → debt payoff)?</i>
Volatility feedback	<i>Do variable-comp months cause outsized asset purchases or forced sales?</i>
Lifestyle-creep radar	<i>Has rising discretionary spend erased gains in asset value?</i>
Wealth-building pacing	<i>At the current savings rate, when will net worth hit ₹ X crore?</i>

Early-retirement Feasibility	<i>If we cap monthly spend at Y, what net-worth milestone brings FIRE forward by N years?</i>
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2. L1 ↔ L3 (Net-Worth × Liquidity)

Theme	Key questions
Asset-liability timing	<i>How many days of runway remain if we prepay the car loan tomorrow?</i>
Liquid vs. illiquid mix	<i>What % of net worth can be liquidated within 7 days?</i>
Drawdown sequencing	<i>If we need ₹10 lakh for IVF, which holdings minimise opportunity-cost and maintain target allocation?</i>
Collateral health	<i>Do property-valuation changes threaten loan-to-value covenants?</i>
Opportunity-fund trigger	<i>At what liquidity buffer does the system ping us to deploy cash into equities?</i>
Fire-sale stress test	<i>If equities fall 30 % and we tap the emergency line, what's the new liquidity ratio?</i>

3. L1 ↔ L4 (Net-Worth × Risk)

Theme	Key questions
Concentration risk	<i>Is > 10 % of net worth tied to employer stock and uninsured?</i>
Insurance sufficiency	<i>After payouts, does net-worth at death meet Aditi's lifetime expense target?</i>
Debt-linked coverage	<i>Does rising home equity necessitate more term insurance to keep dependents whole?</i>
Hedge efficiency	<i>Which liability pre-payment or insurance top-up gives best drop-in overall risk score?</i>
Tail-event simulation	<i>Market crash + medical emergency: what's the residual net worth?</i>
Guarantee vs. self-insurance	<i>Is it cheaper to raise emergency fund by ₹2 lakh or buy critical-illness cover?</i>

4. L1 ↔ L5 (Net-Worth × Tax & Legal)

Theme	Key questions
After-tax true wealth	<i>What is net worth on a post-tax, post-liquidation-cost basis?</i>
Gain-harvest windows	<i>Which asset sales this FY optimise taxable brackets without derailing allocation?</i>
Estate alignment	<i>Are asset titles / nominees consistent with our will's distribution logic?</i>
Ownership structuring	<i>Would shifting rental property into joint names cut tax and balance net-worth attribution?</i>
Exit-planning	<i>How do angel-investment exits affect surcharge thresholds?</i>
Rebalance vs. tax drag	<i>Does rebalancing out of US ETFs this quarter save or cost more after capital-gains tax?</i>

5. L2 ↔ L3 (Income/Expense × Liquidity)

Theme	Key questions
Pay-cycle gaps	<i>Do salaries land before major EMIs, or is overdraft required?</i>
Seasonal crunches	<i>Which month's net spend pushes bank balance below 2-month buffer?</i>
Savings-sweep timing	<i>What sweep schedule (daily vs. monthly) maximises yield without overdraft risk?</i>
Bonus utilisation	<i>If annual bonus is delayed, which planned lump-sum investments get deferred?</i>
Variable income resilience	<i>How many months of expenses are safe if RSU vesting pauses?</i>
Burn-rate alarms	<i>Trigger alert when 30-day burn-rate exceeds Y % of average net income.</i>

6. L2 ↔ L4 (Income/Expense × Risk)

Theme	Key questions
Income-loss buffer	<i>If either of us loses income for 9 months, does insurance + emergency fund cover fixed costs?</i>
Premium affordability	<i>Are insurance premiums staying $\leq Z$ % of gross income?</i>
Expense shock coverage	<i>Which discretionary spends can be cut first to extend emergency runway?</i>
Debt-service stress	<i>Can we handle a 2 % rate hike without crossing 40 % debt-to-income?</i>
Coverage laddering	<i>At what future income level should we top up term cover again?</i>
Income-linked riders	<i>Does adding a waiver-of-premium rider make sense given current spend patterns?</i>

7. L2 ↔ L5 (Income/Expense × Tax)

Theme	Key questions
Deductions vs. cash-flow	<i>How much voluntary NPS fits within surplus without hurting liquidity?</i>
Tax-aware earnings mix	<i>What is the marginal tax rate on freelance income after 80C exhaustion?</i>
Reimbursement optimisation	<i>Would shifting internet/phone bills to salary reimbursements save $\geq ₹X$?</i>
Allowance calibration	<i>Is HRA split optimal between rent paid and taxable salary?</i>
Spend-to-save trade-offs	<i>Does buying a health top-up pay for itself via 80D deductions?</i>
Timing hacks	<i>Should we prepay children's school fees now to claim deduction this FY?</i>

8. L3 ↔ L4 (Liquidity × Risk)

Theme	Key questions
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Emergency-fund adequacy	<i>Post-deductible, how much cash is needed for worst-case hospital bill?</i>
LOC vs. cash	<i>Is it cheaper to keep cash idle or rely on line-of-credit for tail-events?</i>
Collateral call readiness	<i>Do we have enough quick cash to meet margin calls without forced equity sale?</i>
Insurance payout timing	<i>How long between claim event and cash receipt—does liquidity fill that gap?</i>
Layered safety nets	<i>What's the combined coverage period of insurance + emergency fund?</i>
Liquidity stress index	<i>Create a “days-to-distress” metric blending cash, debt covenants, and risk events.</i>

9. L3 ↔ L5 (Liquidity × Tax)

Theme	Key questions
Tax-payment scheduling	<i>Do advance-tax dates align with peak cash months?</i>
Gain-realisation cash impact	<i>Will selling RSUs for tax planning leave bank balance below buffer?</i>
Refund utilisation	<i>Where should anticipated tax refund be parked for short-term yield?</i>
GST/Withholding reserves	<i>Do we segregate enough cash for quarterly GST/TDS obligations on freelance work?</i>
Loan vs. liquidation	<i>Is taking a short-term loan cheaper than liquidating equities (and incurring CGT)?</i>
Tax-efficient cash bucket	<i>Would a 7-day T-bill ladder beat savings-account yield net of tax?</i>

10. L4 ↔ L5 (Risk × Tax & Legal)

Theme	Key questions
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Tax status of payouts	<i>Are all insurance benefits structured to be tax-free under Sec 10D?</i>
Estate liquidity	<i>Will heirs face estate taxes or stamp duty they can't fund without selling assets?</i>
Policy ownership structuring	<i>Whose name should be proposer vs. life-assured to maximise deductions?</i>
Trust vs. nominee	<i>Would a family trust both reduce probate delays and insulate against creditors?</i>
Critical-illness tax relief	<i>Does a lump-sum CI payout jeopardise Section 80D benefits in the same year?</i>
Double-dip opportunities	<i>Can health-insurance top-ups create both risk cover and extra 80D room?</i>

How to operationalise this catalogue

For each bullet above, the system will eventually require:

1. **Cross-layer data joins** – e.g., holdings table × cash-flow calendar.
2. **Derived metrics & thresholds** – net-worth volatility %, debt-to-income, etc.
3. **Scenario engines** – shock simulations, “what-if” tax calculators.
4. **Intent detection** – map user phrasing (“Can we survive a sabbatical?”) to the right query bundle.
5. **Alert/automation hooks** – rules that fire when cross-layer KPIs breach guard-rails.

With this deeper list, you can now draft API contracts, DB schemas, and agent prompts knowing exactly *why* each data relationship matters. Ping me when you're ready to blueprint specific endpoints or calculation modules!